Build-Systeme: Apache Maven

Carsten Gips (HSBI)

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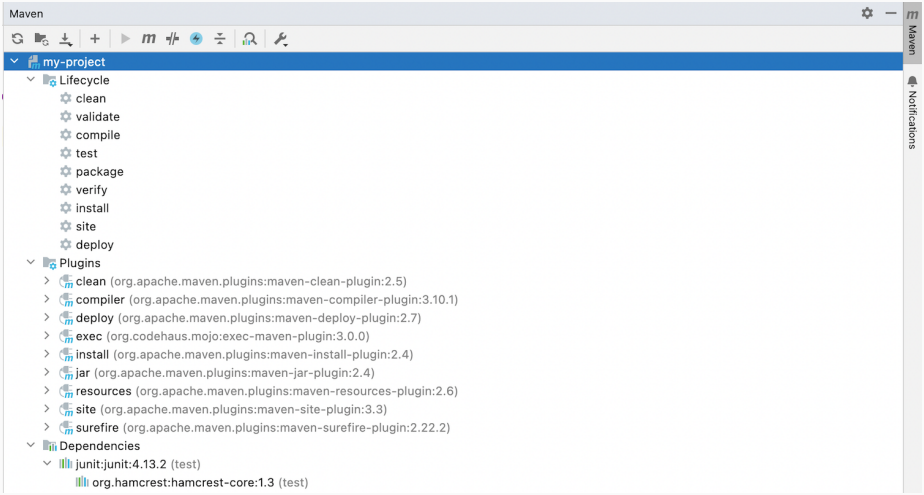
Build-Tool Maven: Alternative zu Ant oder Gradle

```
mvn archetype:generate -DgroupId=de.hsbi.pm -DartifactId=my-project
                       -DarchetypeArtifactId=maven-archetype-quickstart
```

```
my-project
├── pom.xml
└── src
    ├── main
    │   └── java
    │       └── de
    │           └── fhb
    │               └── pm
    │                   └── App.java
    └── test
        └── java
            └── de
                └── fhb
                    └── pm
                        └── AppTest.java
```

Lebenszyklus (eingebaut in Maven)



Project Object Model: *pom.xml*

```xml
<project>
    <!-- aktuelle Version für Maven 2.x-->
    <modelVersion>4.0.0</modelVersion>

    <!-- Basisinformationen -->
    <groupId>de.hsbi.pm</groupId>
    <artifactId>my-project</artifactId>
    <version>1.0-SNAPSHOT</version>

    <!-- Eigenschaften, vergleichbar zu den Properties in Ant -->
    <properties>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
        <maven.compiler.release>17</maven.compiler.release>
    </properties>

    <!-- Abhängigkeiten zu externen Bibliotheken -->
    <dependencies>
        <dependency>
            <groupId>junit</groupId>
            <artifactId>junit</artifactId>
            <version>4.13.2</version>
            <scope>test</scope>
        </dependency>
    </dependencies>
</project>
```

Project Object Model: Plugins

```
<project>
    ...

    <!-- Plugins: Stellen eigene "Goals" zur Verfügung -->
    <build>
        <plugins>
            <plugin>
                <!-- https://mvnrepository.com/artifact/org.apache.maven.plugins/maven-compiler-plugin -->
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-compiler-plugin</artifactId>
                <version>3.10.1</version>
            </plugin>

            <plugin>
                <!-- https://mvnrepository.com/artifact/org.apache.maven.plugins/maven-surefire-plugin -->
                <groupId>org.apache.maven.plugins</groupId>
                <artifactId>maven-surefire-plugin</artifactId>
                <version>2.22.2</version>
            </plugin>
        </plugins>
    </build>
</project>
```

Demo: pom.xml

Wrap-Up

Apache Maven: maven.apache.org, Maven Getting Started Guide

- Automatisieren von Arbeitabläufen
- Apache Maven: Goals, Properties, Dependencies => "*Convention over Configuration*", Java-Standard-Lebenszyklus eingebaut
  - Goals sind auswählbare Ziele, bereitgestellt durch Plugins
  - Abhängigkeiten zwischen Goals möglich
  - Properties agieren wie Variablen, etwa für Versionsnummern
  - Abhängigkeiten zu externen Bibliotheken werden als Dependencies formuliert:

LICENSE

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Exceptions

- Website mvnrepository.com